An FSDS weight on the Q-index

The share names a stale score. The action stays in the table.

1 The index

The control task keeps a visit-bonus table $Q(s, a)$. The state s is the three-way sign of two known reference scores and the step. An earlier table also conditioned on which reference policy held the pseudo-outcome share. That coordinate split the visits and raised the excess on the sign flip. The weight in this note is a term in the decision index. The state is the same two signs and the step.

Four reference policies are scored on every batch, on the same calendar as the scorecard race: the first six rounds against the latest six, refreshed every eight rounds from round 16. The weight uses the two directional scores, the same two whose signs form the state. Write $w_{(1)}$ and $w_{(2)}$ for the larger and the smaller of their shares, and $a_{(1)}(s)$, $a_{(2)}(s)$ for the actions they prescribe. The penalty is the share gap. It falls on the action of the higher share, and only in states where the two actions differ,

$$\delta(s, a) = (w_{(1)} - w_{(2)})\, 1\{a = a_{(1)}(s)\}\, 1\{a_{(1)}(s) \neq a_{(2)}(s)\}.$$

The decision index is

$$Q(s, a) + c(s, a) - \eta\sigma \cdot \frac{H - h}{H} \cdot \delta(s, a),$$

with $\eta \in \{1, 2\}$ and σ the scale of the four reference returns on the first six rounds. The factor $(H-h)/H$ places one episode-level charge along the horizon. Equal shares give $\delta = 0$, so the weight is inert while the two scores still carry the same residual. Agreed actions also give $\delta = 0$: both scores would press the same button, and that button carries no penalty.

The backup is the observed reward plus the max of the next raw table. The share records which reference score still explains the outcome, and it stays out of the value being learned. An unvisited cell still carries the large visit bonus, so the weight cannot hide an action the table has not tried.

One variant sets the weight to zero on a round whose largest share belongs to the zero policy. That is the sign-flip symptom the shut-off was built for: the null score holds the mass, and a penalty aimed at a stale direction has no stale direction left. Across the six pools, the fraction of post-shift rounds on which the zero policy holds the share has median 0.00 on relocation and 0.00 on the flip.

2 Six pools

A reported median is the median, across six pools, of the four-seed mean post-shift excess. A named pool is that four-seed mean, with the seed standard deviation in parentheses. A gap counts only when it exceeds the larger of the two seed standard deviations.

On relocation the menu at $\eta = 2$ is 9.7. Visit-bonus Q with no weight is 15.9. The same table at $\eta = 1$ is 15.7 and at $\eta = 2$ is 15.9. Discounted UCB on the menu is 15.3. The $\eta = 2$ weight is lower than unweighted Q on 0/6 pools, and lower than the menu on 0/6. The menu is lower than

the weight on 2/6. California is 3.3 (0.2) for the menu and 11.7 (0.9) for the weight. Covertype is 7.2 (0.3) for the menu and 26.3 (1.6) for the weight. On AR(1), Gaussian-12, Gaussian-30, and Student-t_5 the gap stays inside the larger seed standard deviation. Early excess, before the ramp, is 12.5 for the unweighted table and 12.5 for the weight. The first-action hit rate is 0.77 for the weight and 0.85 for the menu.

On the sign flip, unweighted visit-bonus Q is 7.4, with first-action hit rate 0.64. The weight at $\eta = 1$ is 10.5 and at $\eta = 2$ is 14.1, with hit rate 0.54. The menu at $\eta = 2$ is 132.0 and discounted UCB is 99.2. Unweighted Q is lower than the $\eta = 2$ weight on 2/6 pools. Gaussian-30 is 12.6 (5.2) for the unweighted table and 25.2 (10.1) for the weight. Student-t_5 is 11.8 (7.1) for the unweighted table and 27.0 (9.3) for the weight. AR(1), California, and Gaussian-12 stay near 3, unweighted table then weight: 2.5 (0.5) and 2.8 (1.4), 2.5 (0.6) and 3.0 (1.5), 3.1 (0.5) and 2.6 (0.8). Covertype is 21.1 (5.6) and 28.5 (11.9); that gap is smaller than the seed standard deviation.

The zero-policy shut-off does not fire once the ramp has crossed one half. On relocation every pool mean matches the always-on weight, so the median stays at 15.9. On the flip the shut-off fires on a few earlier rounds. The post-shift median stays at 14.1. The mean across the six pools moves from 14.84 to 14.82. AR(1) moves from 2.8 (1.4) to 2.7 (1.5). The shut-off is lower than the always-on weight by more than the seed standard deviation on 0/6 pools.

The share names which reference score has stopped explaining the outcome. Once a cell has a value, that name is not the button. Relocation stays with the menu at $\eta = 2$. The sign flip stays with visit-bonus Q, and this weight stays off that table.

Table 1: Median, across six pools, of the four-seed mean post-shift excess return.

scenario	DUCB	UCB-H2	QUCB	QW1	QW2	QW2-Z
relocation	15.3	9.7	15.9	15.7	15.9	15.9
sign flip	99.2	132.0	7.4	10.5	14.1	14.1

Table 2: Relocation. Sum of per-round mean excess return. Seed standard deviation in parentheses.

pool	UCB-H2	QUCB	QW1	QW2	QW2-Z
AR(1)	9.8 (4.8)	10.0 (0.7)	9.8 (0.3)	10.0 (0.4)	10.0 (0.4)
California	3.3 (0.2)	12.3 (0.5)	11.1 (0.4)	11.7 (0.9)	11.7 (0.9)
covertype	7.2 (0.3)	24.0 (1.6)	25.7 (1.1)	26.3 (1.6)	26.3 (1.6)
Gaussian-12	9.5 (2.4)	9.6 (0.6)	9.8 (0.5)	10.4 (0.7)	10.4 (0.7)
Gaussian-30	20.4 (0.6)	19.4 (2.9)	20.5 (1.8)	20.1 (4.6)	20.1 (4.6)
Student-t_5	19.4 (1.3)	20.6 (3.3)	20.3 (3.8)	23.5 (4.7)	23.5 (4.7)

Table 3: Sign flip. Same excess. Seed standard deviation in parentheses.

pool	UCB-H2	QUCB	QW1	QW2	QW2-Z
AR(1)	88.5 (1.6)	2.5 (0.5)	2.6 (1.0)	2.8 (1.4)	2.7 (1.5)
California	78.2 (1.6)	2.5 (0.6)	2.3 (0.8)	3.0 (1.5)	3.0 (1.5)
covertype	160.9 (7.4)	21.1 (5.6)	26.8 (10.7)	28.5 (11.9)	28.5 (11.9)
Gaussian-12	105.5 (3.6)	3.1 (0.5)	2.6 (0.8)	2.6 (0.8)	2.6 (0.8)
Gaussian-30	160.4 (7.1)	12.6 (5.2)	18.4 (5.6)	25.2 (10.1)	25.2 (10.1)
Student-t_5	158.5 (4.9)	11.8 (7.1)	19.8 (5.5)	27.0 (9.3)	27.0 (9.3)

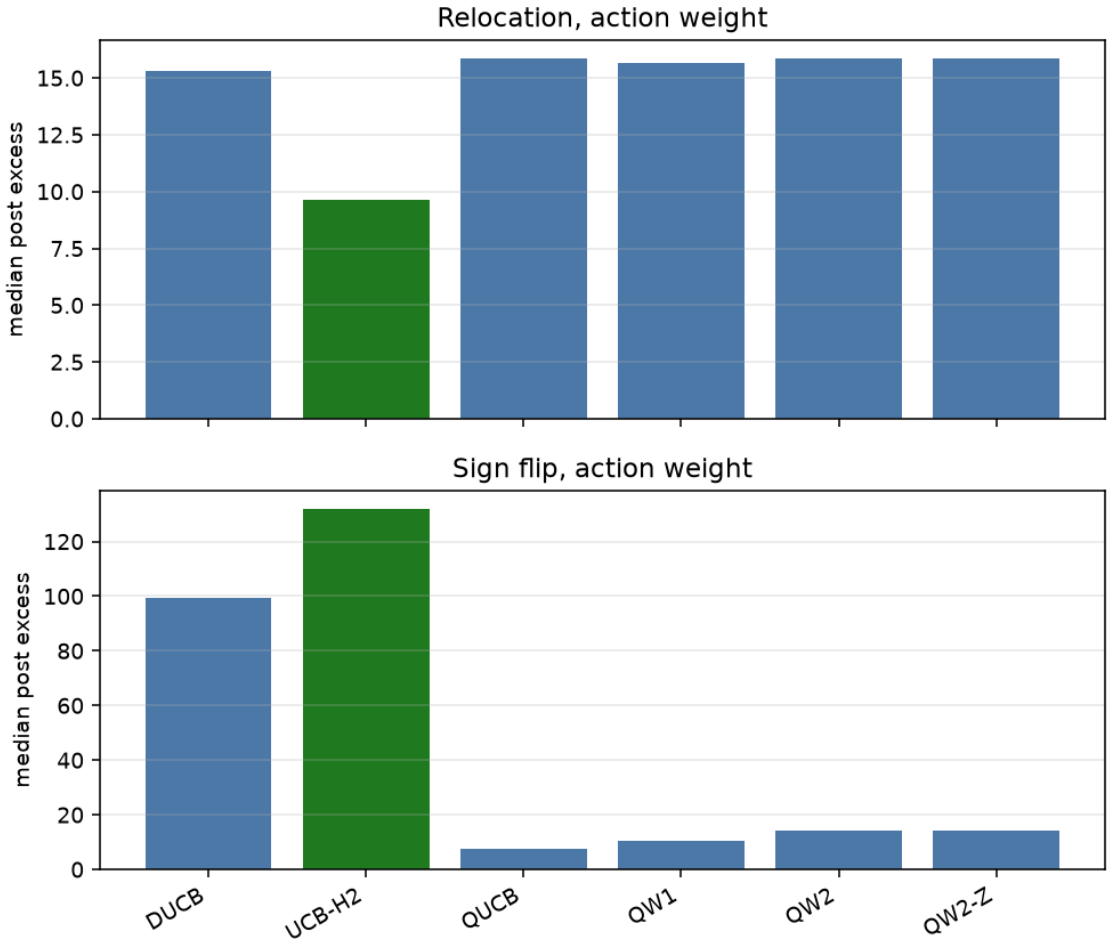


Figure 1: Median post-shift excess across six pools. The menu at $\eta = 2$ is marked in green. QW1 and QW2 subtract the share gap from the visit-bonus index. QW2-Z uses the same index and sets the weight to zero on a round whose largest share is the zero policy.